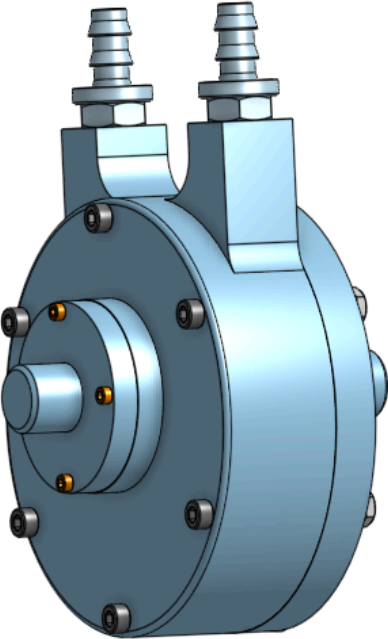


**DRINKING WATER PUMP**

**ASSEMBLY INSTRUCTIONS**  
**DOC-000001**

|                                                                                     |                                            |          |            |
|-------------------------------------------------------------------------------------|--------------------------------------------|----------|------------|
| DOCUMENT NUMBER:                                                                    | DOC-000001                                 | REVISION | 00         |
| DOCUMENT NAME:                                                                      | ASSEMBLY INSTRUCTIONS, DRINKING WATER PUMP |          |            |
| ASSEMBLY REFERENCED:                                                                | DWP-000001                                 | REVISION | 00         |
| CURRENT REVISION WRITTEN BY                                                         | BCS                                        | DATE     | 12/15/2024 |
| CURRENT REVISION CHECKED BY                                                         | [initials]                                 | DATE     | 2024-      |
| CURRENT REVISION APPROVED BY                                                        | [initials]                                 | DATE     | 2024-      |
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**DOCUMENT REVISION HISTORY**

| Revision | Details                     | Written By | ECO Number | Date       |
|----------|-----------------------------|------------|------------|------------|
| 00       | Initial Release of Document | BCS        | N/A        | 12/15/2024 |

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**1. RISKS AND HAZARDS**

- Ensure the operator is properly trained on assembly best practices and safety precautions.

**2. EQUIPMENT OR FIXTURES REQUIRED**

| Quantity | Description                       |
|----------|-----------------------------------|
| 1        | Arbor Press and Bearing Depth Jig |
| 1        | 5/32" Hex Key                     |
| 1        | 3/8" Combination Wrench           |

**3. CONSUMABLES REQUIRED**

| Quantity | Description                                      |
|----------|--------------------------------------------------|
| ~1 tbsp  | CRC Multi-Purpose Food Grade Grease              |
| ~0.5 oz  | Threadlocker Loctite 243, Food Safe Threadlocker |

**4. REFERENCED DOCUMENTS**

| Document number | Description                          |
|-----------------|--------------------------------------|
| DWP-000001      | DRINKING WATER PUMP ASSEMBLY AND BOM |

**5. ENVIRONMENT**

- Ensure that all assembly takes place in a clean area that is free from contaminants.
- Ensure assembly is done in a food safe environment in accordance with NSF/ANSI 2 sanitation requirements. Appropriate PPE must be worn.

**6. BUILD DURATION**

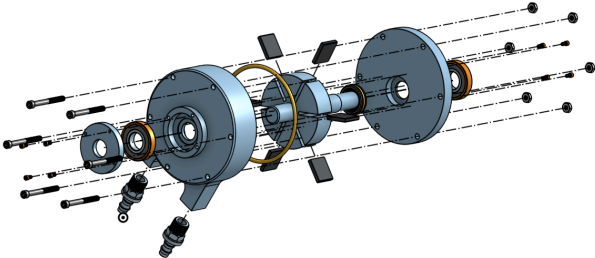
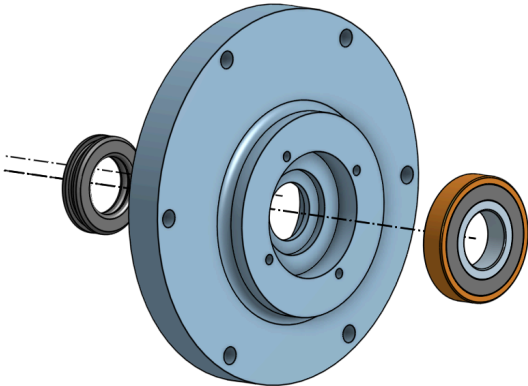
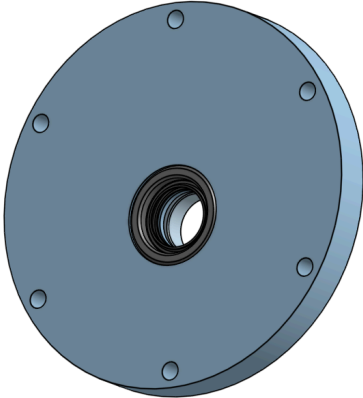
- The total build duration of this assembly should be 15 minutes per unit.

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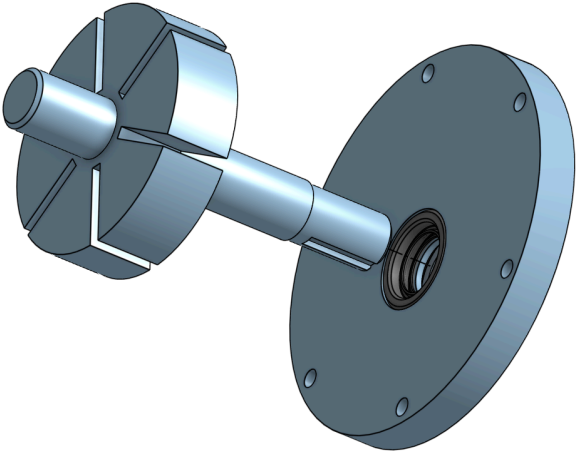
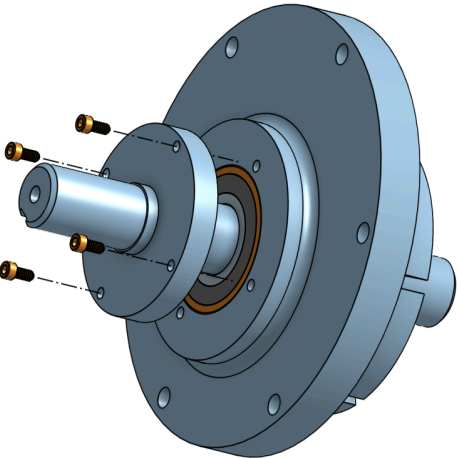
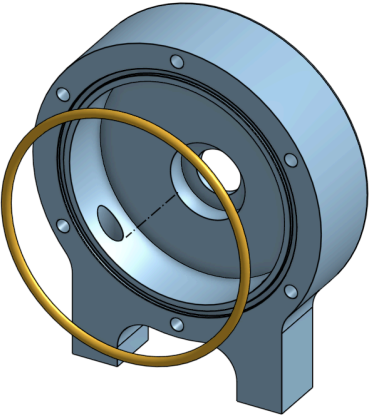
7. ASSEMBLY PROCEDURE

| STEP NO | DESCRIPTION                                                                                                                           | Illustration                                                                          |
|---------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 01      | Ensure all parts are to spec and have passed incoming quality procedures.                                                             |     |
| 02      | Insert Permanently Lubricated Bearing (8) into the outside of Front Plate (6). Insert Shaft Seal (8) on opposite side of Front Plate. |    |
| 03      | Apply CRC Multi-Purpose Food Grade Grease to the inside of the Shaft Seal (8). Do not use more than necessary.                        |  |

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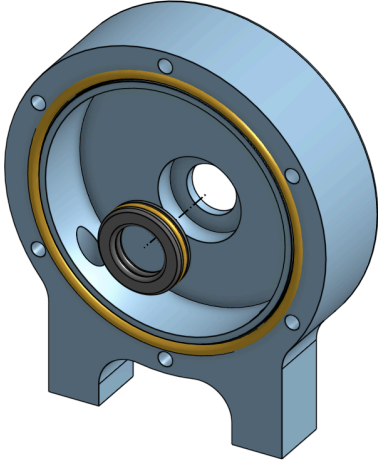
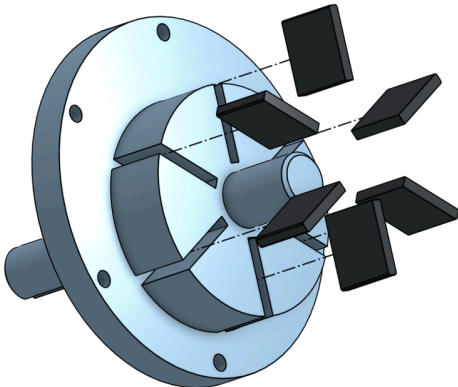
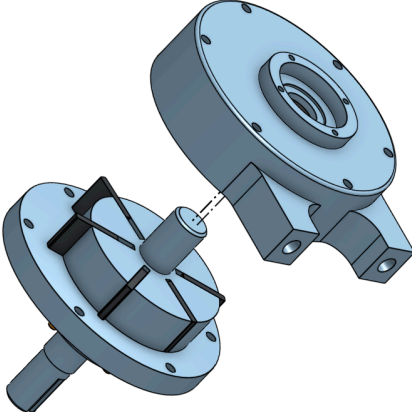
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|    |                                                                                                                                       |                                                                                      |
|----|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 04 | With the bearing facing down, use the arbor press to press the Rotor (10) into the bearing. Press until motion is stopped by the jig. |    |
| 05 | Attach Bearing Housing Cover (4) to the Front Plate (6) using 4X #4 Nickel-Alloy Socket Head Screw (5). Hand tighten.                 |   |
| 06 | Insert HNBR O-Ring into the groove on the front side of Pump Housing (1).                                                             |  |

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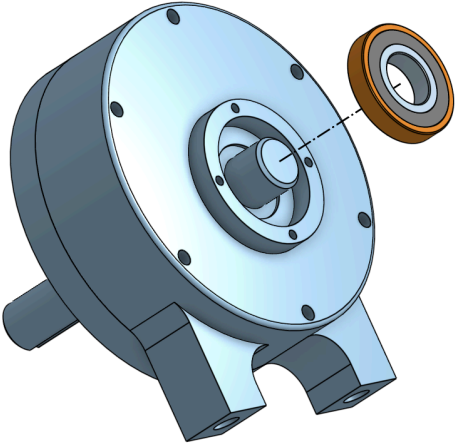
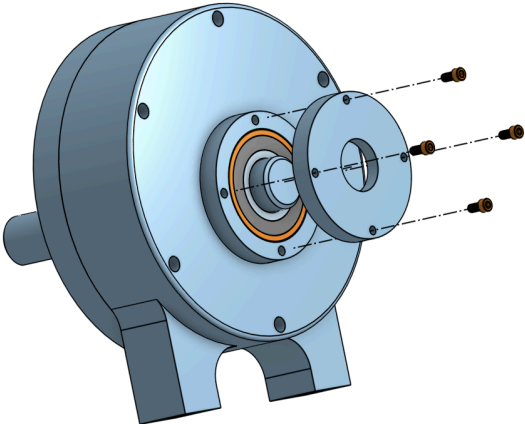
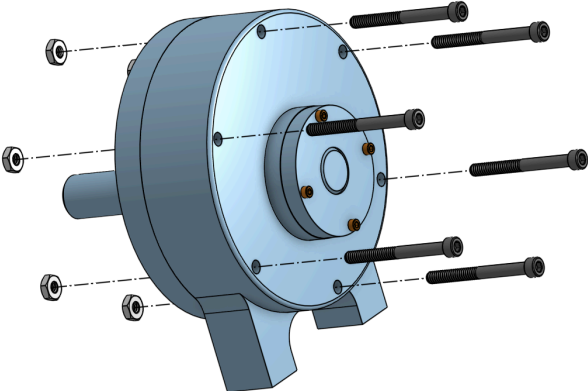
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|    |                                                                                                                                                                                                                                                                              |                                                                                       |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 07 | Insert Shaft Seal (8) into the inside of Pump Housing (1). Apply grease to the inside of Shaft Seal without using more than necessary.                                                                                                                                       |    |
| 08 | With the Rotor and Front Plate oriented vertically with the Rotor on top, slide 6X Carbon Graphite Vane (3) into the slots of the rotor such that the vanes sit on the surface of the Front Plate. Ensure the chamfered edges point away from the central axis of the rotor. |    |
| 07 | Lower the Pump Housing such that the axis of the rotor goes through the Shaft Seal. Push until Front Plate and Housing are touching.                                                                                                                                         |  |

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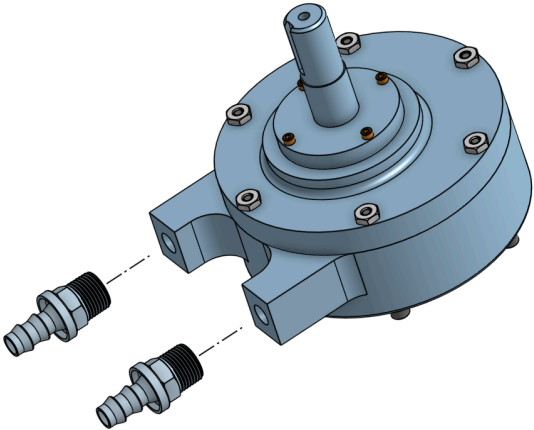
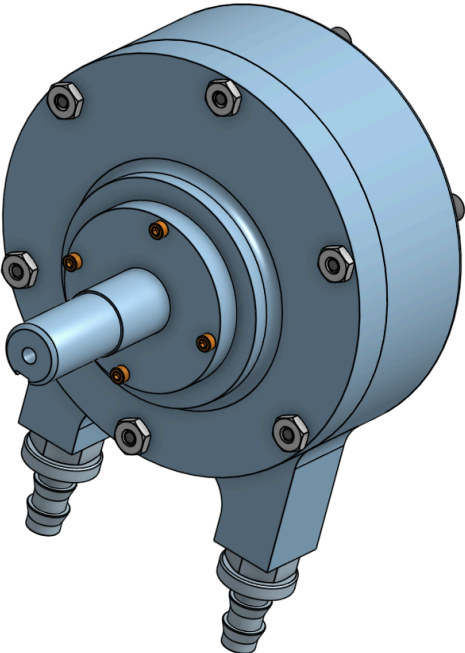
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|    |                                                                                                                                                                                                                                           |                                                                                                                                                                                                               |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 08 | Slide the final Permanently Lubricated Bearing (9) onto the back of the rotor axis.                                                                                                                                                       |                                                                                                                             |
| 09 | Cover the rear bearing with the final Bearing Housing Cover (4). Fasten using 4X #4 Nickel-Alloy Socket Head Screw (5). Hand Tighten.                                                                                                     |                                                                                                                            |
| 10 | Fasten the Front Plate to the Housing using 6X #10-24 X 2" Zinc-Plated Alloy Steel Bolts (7) and 6X Super Corrosion Resistant Hex Nuts (2). Apply a small amount of Threadlocker Loctite 243 to each bolt before fastening. Hand Tighten. |  |

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|    |                                                                                                                                             |                                                                                     |
|----|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 11 | Screw in the two 3/8"x3/8" Brass Hose Fittings (12) into the inlet and the outlet of the assembled pump, use a spanner to tighten fittings. |   |
| 12 | <b>Assembly Complete</b> – Store the module in a clean container and queue for assembly test and calibration.                               |  |

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